RELATIONAL ALGEBRA

Learning Objective :

Use relational algebra and calculus in retrieving data from the database systems.

Relational Query Languages

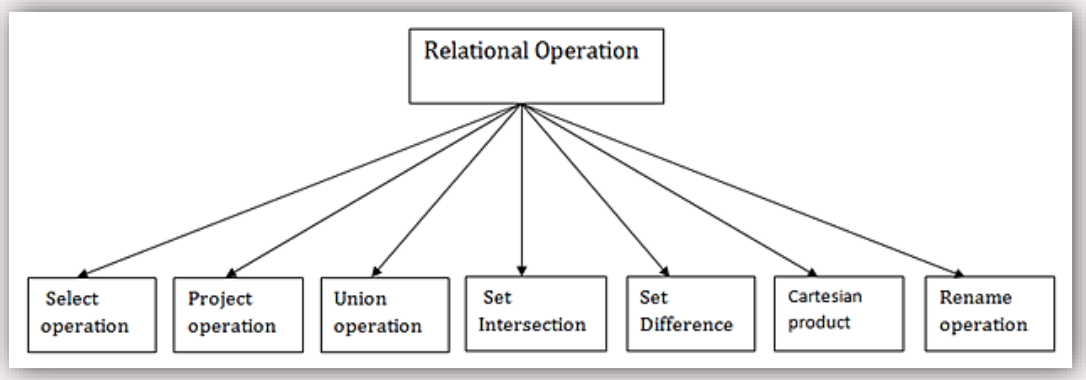
- **Query Languages** – allow manipulation and retrieval of data from a database.
- 2 mathematical Query Languages form the basis for “real” languages (e.g. SQL), and for implementation:
  - ✓ Relational Algebra: More operational, very useful for representing execution plans.
  - ✓ Relational Calculus: Let users describe what they want, rather than how to compute it.

Relational Algebra

- A procedural query language, which takes instances of relations as input and yields instances of relations as output.
- It uses operators to perform queries.
  - An operator can be either unary or binary.
- Relational algebra is performed recursively on a relation and intermediate results are also considered relations.

Operations of Relational Algebra

- Select (σ)
- Project (∏)
- Union (∪)
- Set Intersection (∩)
- Set difference (−)
- Cartesian product (x)
- Rename (ρ)



Select (σ)

- It selects tuples that satisfy the given predicate from a relation.
- Notation: $\sigma_p(r)$
  - **σ** stands for selection predicate
  - **r** stands for relation
  - *p* is prepositional logic formula which may use connectors like **and, or,** and **not**

  These terms may use relational operators like: =, ≠, ≥, < , >, ≤.

Select (σ) : Example

- $\sigma_{\text{subject = "database"}}(\text{Books})$
  - *Output*: Selects tuples from books where subject is 'database'.
- $\sigma_{\text{subject = "database" and price = "450"}}(\text{Books})$
  - *Output*: Selects tuples from books where subject is 'database' and 'price' is 450.
- $\sigma_{\text{subject = "database" and price = "450" or year > "2010"}}(\text{Books})$
  - *Output*: Selects tuples from books where subject is 'database' and 'price' is 450 or those books published after 2010.

TASK 1 : Select (σ)

A

B

LOAN

BRANCH_NAME	LOAN_NO	AMOUNT
Downtown	L-17	1000
Redwood	L-23	2000
Perryride	L-15	1500
Downtown	L-14	1500
Mianus	L-13	500
Roundhill	L-11	900
Perryride	L-16	1300

Project Operation (∏)

- It projects column(s) that satisfy a given predicate.
- Notation: $\prod_{A1,\ A2,\ An}(r)$
  - A1, A2 , An are attribute names of relation r
- Duplicate rows are automatically eliminated, as relation is a set.
- Example:

  $\prod_{subject,\ author}(Books)$

  - Output: Selects and projects columns named as subject and author from the relation Books.

Project Operation (∏) : Example

- $\Pi_{\text{subject, author}}$ (Books)
  - Output: Selects and projects columns named as subject and author from the relation Books.
- $\Pi_{\text{CustomerName, Status}}$ (Customers)

CustomerID	CustomerName	Status
1	Google	Active
2	Amazon	Active
3	Apple	Inactive
4	Alibaba	Active

CustomerName	Status
Google	Active
Amazon	Active
Apple	Inactive
Alibaba	Active

TASK 2 : Project Operation (∏)

A

B

EMPLOYEE

NAME	ADDRESS	DESIGNATION
Marian	Laguna	Programmer
Jason	Davao	System Analyst
Bernard	Baguio	UX Designer
Lian	Baybay	Database Administrator
Rosa	Ilocos	IT Specialist
Gil	Manila	AI Engineer
Peter	Tagaytay	Computer Engineer

Union Operation (∪)

- It performs binary union between two given relations and is defined as

$$r \cup s = \{ t \mid t \in r \text{ or } t \in s\}$$

- Notation: $r \cup s$
  - r and s are either database relations or relation result set (temporary relation)
- For a union operation to be valid, the following conditions must hold –
  - r, and s must have the same number of attributes.
  - Attribute domains must be compatible.
  - Duplicate tuples are automatically eliminated.

Union Operation (∪) : Example

- Π_{author} (Books) ∪ Π_{author} (Articles)
  - Output: Projects the names of the authors who have either written a book or an article or both.
- A ∪ B

Table A		Table B	
column 1	column 2	column 1	column 2
1	1	1	1
1	2	1	3

Table A ∪ B	
column 1	column 2
1	1
1	2
1	3

TASK 3 : Union Operation (∪)

A

NAME
Marian
Gil
Jason
Bernard
May
Lian
Peter
William

B

DEPOSITOR

NAME	ACCOUNT_NO
Marian	A-101
Jason	A-121
Bernard	A-321
Lian	A-176
May	A-273
Gil	A-472
Peter	A-284

BORROWER

NAME	ACCOUNT_NO
Gil	L-17
Jason	L-23
May	L-15
Peter	L-14
William	L-93
Gil	L-17

Intersection (∩)

- Defines a relation consisting of a set of all tuple that are in both A and B. However, A and B must be union-compatible.
- Example: A ∩ B

Table A		Table B	
column 1	column 2	column 1	column 2
1	1	1	1
1	2	1	3

Table A ∩ B	
column 1	column 2
1	1

TASK 4 : Intersection (∩)

A

NAME
Jason
May
Gil
Peter

B

DEPOSITOR

NAME	ACCOUNT_NO
Marian	A-101
Jason	A-121
Bernard	A-321
Lian	A-176
May	A-273
Gil	A-472
Peter	A-284

BORROWER

NAME	ACCOUNT_NO
Gil	L-17
Jason	L-23
May	L-15
Peter	L-14
William	L-93
Gil	L-17

Set Difference (-)

- Notation: A - B
- The result of A – B, is a relation which includes all tuples that are in A but not in B.
  - The attribute name of A has to match with the attribute name in B.
  - The two-operand relations A and B should be either compatible or Union compatible.
  - It should be defined relation consisting of the tuples that are in relation A, but not in B.
- Example :

Table A		Table B	
column 1	column 2	column 1	column 2
1	1	1	1
1	2	1	3

Table A – B	
column 1	column 2
1	2

TASK 5 : Set Difference (-)

A B

Name
William

DEPOSITOR

NAME	ACCOUNT_NO
Marian	A-101
Jason	A-121
Bernard	A-321
Lian	A-176
May	A-273
Gil	A-472
Peter	A-284

BORROWER

NAME	ACCOUNT_NO
Gil	L-17
Jason	L-23
May	L-15
Peter	L-14
William	L-93

Cartesian Product (X)

- Combines information of two different relations into one.
- Notation: r x s
  - r and s are relations and their output will be defined as –

$$r \times s = \{ q\ t \mid q \in r \text{ and } t \in s \}$$

- Example:

$$\sigma_{\text{author = 'tutorialspoint'}}(\text{Books } \mathbf{X} \text{ Articles})$$

  - Output : Yields a relation, which shows all the books and articles written by tutorialspoint.

TASK 6 : Cartesian Product (X)

A

B

EMP_ID	EMP_NAME	EMP_DEPT	DEPT_ID	DEPT_NAME
1	Smith	A	A	Comp. Sci.
1	Smith	A	B	Statistics
1	Smith	A	C	Arts
2	Harry	C	A	Comp. Sci.
2	Harry	C	B	Statistics
2	Harry	C	C	Arts
3	John	B	A	Comp. Sci.
3	John	B	B	Statistics
3	John	B	C	Arts

EMPLOYEE

EMP_ID	EMP_NAME	EMP_DEPT
1	Smith	A
2	Harry	C
3	John	B

DEPARTMENT

DEPT_ID	DEPT_NAME
A	Comp. Sci.
B	Statistics
C	Arts

Rename Operation (ρ)

- The results of relational algebra are also relations but without any name. The rename operation allows us to rename the output relation. 'rename' operation is denoted with small Greek letter rho ρ
- Notation: ρ x (E)
  - the result of expression E is saved with name of x.
- ρ(newName , oldName)

JOIN OPERATIONS

- Join is a combination of a Cartesian product followed by a selection process.
- A Join operation pairs two tuples from different relations, if and only if a given join condition is satisfied.

- Inner Join
  - Theta join
  - EQUI join
  - Natural join
- Outer Join:
  - Left outer join
  - Right outer join
  - Full outer join

Inner Join

- An inner join includes only those tuples with matching attributes and the rest are discarded in the resulting relation.
- Types:
  - Theta join
  - EQUI join
  - Natural join

Inner Join : Theta Join (θ)

- Theta Join (θ) combines tuples from different relations provided they satisfy the theta condition.
- Notation: $R1 \bowtie_{\theta} R2$
  - R1 and R2 are relations having attributes (A1, A2, .., An) and (B1, B2,.. ,Bn) such that the attributes don't have anything in common, that is $R1 \cap R2 = \Phi$.
- Theta join can use all kinds of comparison operators.

Student		
SID	**Name**	**Std**
101	Alex	10
102	Maria	11

Subjects	
Class	**Subject**
10	Math
10	English
11	Music
11	Sports

Student_detail				
SID	**Name**	**Std**	**Class**	**Subject**
101	Alex	10	10	Math
101	Alex	10	10	English
102	Maria	11	11	Music
102	Maria	11	11	Sports

$$\text{Student_ Detail} - \text{STUDENT} \bowtie_{\text{Student.Std = Subjects.Class}} \text{SUBJECTS}$$

Inner Join : Equijoin

- When Theta join uses only equality comparison operator, it is said to be equijoin.

Inner Join : Natural Join (⋈)

- Natural join does not use any comparison operator.
- It does not concatenate the way a Cartesian product does.
- Perform a Natural Join only if there is at least one common attribute that exists between two relations
  - the attributes must have the same name and domain
- Natural join acts on those matching attributes where the values of attributes in both the relations are same

Inner Join : Natural Join (⋈)

Courses		
CID	**Course**	**Dept**
CS01	Database	CS
ME01	Mechanics	ME
EE01	Electronics	EE

HoD	
Dept	**Head**
CS	Alex
ME	Maya
EE	Mira

Courses ⋈ HoD			
Dept	**CID**	**Course**	**Head**
CS	CS01	Database	Alex
ME	ME01	Mechanics	Maya
EE	EE01	Electronics	Mira

TASK 7 : Natural Join

A

B

EMP_NAME	SALARY
Smith	50000
Harry	30000
John	25000

EMPLOYEE

EMP_CODE	EMP_NAME
101	Smith
102	Harry
103	John

SALARY

EMP_CODE	SALARY
101	50000
102	30000
103	25000

Outer Join

- The outer join operation is an extension of the join operation.
- It is used to deal with missing information.
- Types:
  - Left outer join
  - Right outer join
  - Full outer join

Left Outer Join (R ⟕ S)

- All the tuples from the Left relation, R, are included in the resulting relation.
- If there are tuples in R without any matching tuple in the Right relation S, then the S-attributes of the resulting relation are made NULL..

TASK 8 : Left Outer Join

A (EMPLOYEE ⋈ FACT_WORKERS)

B (FACT_WORKERS ⋈ EMPLOYEE)

EMPLOYEE

EMP_NAME	STREET	CITY
Ram	Rizal	Manila
Shen	Zone 4	Baybay
Ruby	San Isidro	Biliran
Harry	Ipil	Ormoc

FACT_WORKERS

EMP_NAME	BRANCH	SALARY
Ram	Infosys	10000
Shen	Wipro	20000
Koby	HCL	30000
Harry	TCS	50000

Left Outer Join (R ⟕ S)

Left	
A	**B**
100	Database
101	Mechanics
102	Electronics

Right	
A	**B**
100	Alex
102	Maya
104	Mira

Courses ⟕ HoD			
A	**B**	**C**	**D**
100	Database	100	Alex
101	Mechanics	---	---
102	Electronics	102	Maya

Right Outer Join (R ⟖ S)

- All the tuples from the Right relation, S, are included in the resulting relation.
- If there are tuples in S without any matching tuple in R, then the R-attributes of resulting relation are made NULL.

Right Outer Join (R ⟖ S)

Left	
A	B
100	Database
101	Mechanics
102	Electronics

Right	
A	B
100	Alex
102	Maya
104	Mira

Courses ⟖ HoD			
A	B	C	D
100	Database	100	Alex
102	Electronics	102	Maya
---	---	104	Mira

Full Outer Join (R ⟗ S)

- All the tuples from both participating relations are included in the resulting relation.
- If there are no matching tuples for both relations, their respective unmatched attributes are made NULL.

Full Outer Join (R ⟗ S)

Left	
A	**B**
100	Database
101	Mechanics
102	Electronics

Right	
A	**B**
100	Alex
102	Maya
104	Mira

Courses ⟗ HoD			
A	**B**	**C**	**D**
100	Database	100	Alex
101	Mechanics	---	---
102	Electronics	102	Maya
---	---	104	Mira

Right Outer Join (R ⟖ S)

- All the tuples from the Right relation, S, are included in the resulting relation.
- If there are tuples in S without any matching tuple in R, then the R-attributes of resulting relation are made NULL.

References:

- https://www.guru99.com/relational-algebra-dbms.html
- https://www.tutorialspoint.com/dbms/database_joins.htm#:~:text=Join%20is%20a%20combination%20of,types%20in%20the%20following%20sections.
- https://www.javatpoint.com/dbms-relational-algebra